

Statistics Teaching System: Architecture and Operations Guide

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1 System Overview

This document describes the complete architecture and operational procedures of the **Statistics Teaching System (stat)**, designed for the English Taught Program in International Business (ETP) at Shih Chien University. The system supports two sequential courses—*Statistics (1)* and *Statistics (2)*—and is built on a core technology stack of R, Quarto, and R Markdown, integrating website publication, in-class exercises, and automated grading into a unified workflow.

Course website: <https://yyliou.github.io/stat> **Instructor:** Yu-You Liou (d10627008@ntu.edu.tw)

1.1 Course Structure

Table 1: Course Identification and Scheduling

Course	Code	Schedule	Room	Chapters
Statistics (1)	EIB-21E-01-A1	Tuesday, Periods 8-9 (15:10-17:00)	L309	Ch. 1-7
Statistics (2)	EIB-21F-01-A1	Thursday, Periods 6-7 (13:10-15:00)	L309	Ch. 8-14

Each session is structured as two consecutive hours: the first hour is a **Lecture** covering theoretical foundations, and the second hour is a **Lab** in which students apply concepts through in-class exercises using the R programming language.

1.2 Assessment Structure

Table 2: Assessment Weighting and Components

Component	Weight	Details
In-Class Exercises	36%	12 exercises x 3% each
Midterm Examination	32%	
Final Examination	32%	

2 Directory Architecture

2.1 Root Directory Structure

The system root (`stat/`) is organized as follows:

```
stat/
|-- _quarto.yml          # Quarto website global configuration
|-- web.Rproj           # RStudio project file
|-- styles.css          # Custom CSS stylesheet
|-- index.qmd           # Homepage (course overview, examination regulations)
|-- sylla.qmd           # Syllabus landing page (links to PDF syllabi)
|-- slide.qmd           # Slide index page
|-- prob.qmd            # Problem set page (submission requirements)
|-- r.qmd               # R installation and usage guide
|
|-- ch2/ ... ch14/      # Per-chapter lecture slide directories
|   |-- chN.qmd         # Quarto Reveal.js slide source for chapter N
|
|-- stat1-ps/           # Statistics (1) problem sets
|   |-- ex1.qmd         # Exercise 1 source (Reveal.js format)
|   |-- ex1.html        # Compiled HTML for publication
|
|-- stat2-ps/           # Statistics (2) problem sets (same structure)
|
|-- stat1-ex/           # Statistics (1) student submissions
|   |-- ex1/
|       |-- ex1-[StudentID].pdf
|   |-- ex2/ ... ex12/
|
|-- stat2-ex/           # Statistics (2) student submissions (same structure)
|
|-- stu1/               # Statistics (1) roster and grading outputs
|   |-- [timestamp].xls # Official roster exported from university system
|   |-- ex[N]/
|       |-- ex-score.xlsx
|       |-- stat.Rmd
|       |-- stat.pdf
|
|-- stu2/               # Statistics (2) roster and grading outputs (same structure)
|
|-- sylla/              # Syllabus source files
|   |-- stat1.Rmd / stat1.html
|   |-- stat2.Rmd / stat2.html
|
|-- prompt/             # AI-assisted prompt documents
|   |-- prompt.Rmd      # Slide generation prompt (instructor use)
|   |-- ex-review1.Rmd  # Grading prompt for Statistics (1)
|   |-- ex-review2.Rmd  # Grading prompt for Statistics (2)
|   |-- stat-system-guide.Rmd # This document
```

2.2 Design Principles

The directory structure adheres to four organizational principles:

1. **One directory per chapter.** Directories `ch2/` through `ch14/` are self-contained units, facilitating independent version control and selective rendering.
2. **Separation of problem sets and submissions.** `stat1-ps/` (instructor-authored questions) and `stat1-ex/` (student submissions) maintain distinct responsibilities and access patterns.

3. **Co-location of rosters and grading outputs.** Each `stu2/` or `stu1/` directory holds both the official class roster and all automated grading artifacts, enabling single-location lookup.
4. **Centralized prompt management.** All AI-assisted documents are isolated within `prompt/`, decoupled from pedagogical content.

3 Technology Stack

3.1 Core Tools

Table 3: Core Software Components

Tool	Minimum Version	Purpose
R	≥ 4.2	Statistical computation and document generation
RStudio Desktop	Latest stable	Integrated development environment
Quarto	≥ 1.3	Website and slide publication
R Markdown	≥ 2.20	Assignment templates and grading reports
TinyTeX	Current	LaTeX engine for PDF output

3.2 Required R Packages

The following packages must be installed prior to system operation:

```
# Document generation
install.packages(c("rmarkdown", "knitr", "tinytex"))
tinytex::install_tinytex()

# Data manipulation and visualization
install.packages(c("tidyverse", "ggplot2", "patchwork"))

# Spreadsheet I/O for grading reports
install.packages(c("readxl", "openxlsx"))

# Academic integrity (UUID generation)
install.packages("uuid")
```

3.3 Website Publication Architecture

The course website is built as a **Quarto Website** project, governed by `_quarto.yml`. Key configuration decisions are summarized below.

Table 4: Quarto Website Configuration Summary

Setting	Value
output-dir	. (HTML output at repository root for GitHub Pages)
Theme	Minty + custom brand CSS
Navigation bar	Syllabus -> Slides -> Problem Sets -> R
Slide format	Reveal.js
Math rendering	KaTeX

4 Lecture Slide Authoring Workflow

4.1 File Format

Each chapter's slides reside in `chN/chN.qmd` with the following YAML header:

```
---
title: "Statistics 1"
subtitle: "Chapter N: [Title]"
author: "Yu-You Liou"
date: "2026-06-02"
format:
  revealjs:
    html-math-method: katex
theme: serif
footer: "Elementary Statistics: A Step by Step Approach"
editor: visual
---
```

4.2 Authoring Standards

All slide content must conform to the following conventions:

- Each slide begins with a level-2 heading (`##`).
- Mathematical expressions use standard LaTeX syntax, e.g., $\bar{x} = \frac{\sum x}{n}$.
- Definitions and key formulae are enclosed in Quarto callout blocks:

```
::: {.callout-note}
**Definition** ...
:::
```

- Every textbook example provides two parallel solutions: (1) a complete algebraic derivation in LaTeX, and (2) an executable R implementation with displayed output.
- All prose must be original; verbatim reproduction of textbook text is prohibited.

4.3 Generating a New Chapter

Step 1. Duplicate the previous chapter directory:

```
cp -r ch2/ ch3/
```

Step 2. Use the prompt in `prompt/prompt.Rmd` with an AI assistant

to generate content from the corresponding textbook chapter.

Step 3. Render in RStudio:

```
quarto render ch3/ch3.qmd
```

Step 4. Update the chapter index in `slide.qmd`.

5 Problem Set Management

5.1 Directory Mapping

Table 5: Problem Set and Submission Directory Mapping

Course	Problem Set Directory	Submission Directory
Statistics (1)	stat1-ps/	stat1-ex/
Statistics (2)	stat2-ps/	stat2-ex/

5.2 Problem Set Format

Problem sets are authored as Reveal.js presentations (`stat1-ps/exN.qmd`), with one question per slide. A representative header:

```
---
title: "Statistics 1"
subtitle: "In-Class Exercise N"
format:
  revealjs:
    html-math-method: katex
---
```

5.3 Submission Requirements

Students must use the official template below and submit to `stat1-ex/exN/` using the exact naming convention `exN-[StudentID].pdf`.

```
---
title: "Ex. N"
author: "StudentID"
date: "2026-06-02"
output: pdf_document
---
```

```
` ` `{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
` ` `

` ` `{r}
print(uuid::UUIDgenerate())
` ` `
```

Every submission must include the UUID generation code block. This serves as the primary mechanism for academic integrity verification.

6 Automated Grading System

6.1 Process Overview

The grading pipeline consists of four sequential stages:

```

STAGE 1 | Initialization and Roster Reconciliation
  Read stu[N]/[timestamp].xls -> construct student roster
  Cross-reference stat[N]-ex/exN/ -> flag absent students
  as Missing (Score = 0)

STAGE 2 | Compliance Filtering (sequential; stops at first violation)
  (i) Submission timestamp > deadline (UTC+8)
      -> Late Submission (Score = 0)
  (ii) File is not PDF or does not match exN-[StudentID].pdf
      -> Incorrect File Name/Format (Score = 0)
  (iii) YAML missing title/author/date, or UUID chunk absent,
        or author field does not match submitting student ID
      -> Incorrect Internal Format (Score = 0)
  (iv) UUID value identical to another submission
      -> Plagiarism (Score = 0; all implicated students)

STAGE 3 | Academic Content Grading
  Total: 100 points, allocated equally across N questions
  Each question scored against stat[N]-ps/exN.qmd answer key
  Incorrect items noted as: [QN] is incorrect

STAGE 4 | Output Generation
  stu[N]/exN/ex-score.xlsx <- Scoreboard (4 columns)
  stu[N]/exN/stat.Rmd      <- Analytics report source
  stu[N]/exN/stat.pdf      <- Compiled analytics report

```

6.2 Scoreboard Format (ex-score.xlsx)

Table 6: Scoreboard Column Specification

Column	Type	Description
StudentID	Character	University student identification number
Name	Character	Student full name as listed in official roster
Score	Integer (0-100)	Final score awarded for the submission
Note	Character	Compliance violation code or incorrect question numbers

6.3 Analytics Report (stat.Rmd)

The analytics report reads `ex-score.xlsx` and produces three outputs:

1. **Descriptive statistics table:** minimum, Q_1 , median, Q_3 , maximum, and mean of the score distribution.
2. **Boxplot:** a publication-quality visualization of the score distribution rendered via `ggplot2`.
3. **Submission summary table:** counts for Total, Missing, Late Submission, Incorrect Format, and Plagiarism.

6.4 Grading Note Codes

Table 7: Grading Note Code Reference

Code	Description	Score
Missing	No file submitted by the deadline	0
Late Submission	File submitted after the official deadline	0
Incorrect File Name/Format	File is not PDF or does not follow naming convention exN-[ID].pdf	0
Incorrect Internal Format	YAML header incomplete, author ID mismatch, or UUID chunk absent	0
Plagiarism	UUID value matches that of one or more other submissions	0
[QN] is incorrect	Question N answer is incorrect or methodology is flawed	Partial deduction

7 Academic Integrity Mechanisms

7.1 UUID Verification

Every submission must execute and display output from the following R code block:

```
print(uuid::UUIDgenerate())
```

`uuid::UUIDgenerate()` produces a Version 4 UUID (Universally Unique Identifier) on each invocation, e.g., 6d2b9fda-2d9b-48f3-84bc-f82eabc39b4d. The grading system compares this value across all submissions for a given exercise. If two or more files share an identical UUID, all implicated students receive a score of zero with the note **Plagiarism**.

Important implementation notes:

- The compliance filter is applied sequentially per file. A file that fails Stage 2 check (iii) (Incorrect Internal Format) is not evaluated for plagiarism within its own record; however, its UUID remains in the comparison pool and may trigger a Plagiarism flag in other students' records.
- The `author` field in the YAML header must contain the submitting student's own ID. Use of another student's ID constitutes a format violation independent of the plagiarism check.

7.2 Examination Regulations

Table 8: Examination Conduct Regulations

Regulation	Policy
Mandatory attendance	Roll call 5 minutes before start; late arrival or absence results in score = 0
Network restriction	System tracks IP addresses; campus Wi-Fi required; non-campus IP results in score = 0
Authorized resources	Open-book; reference materials and AI tools are permitted as auxiliary aids
Media prohibition	Photography, video/audio recording, and screenshotting result in score = 0
Communication prohibition	Interpersonal communication and disruptive behavior result in score = 0
Copy-paste disabled	Copy-paste functionality is disabled by the examination system

7.3 AI Use Policy

Students may use AI tools (e.g., ChatGPT, Gemini) as learning aids. However, students must be capable of explaining and defending any code or analysis they submit. Uncritical submission of AI-generated content without demonstrated understanding constitutes academic dishonesty and is subject to the same consequences as plagiarism.

8 AI Prompt Management

The `prompt/` directory contains structured prompt documents for AI-assisted course material generation and grading.

Table 9: Prompt Directory File Inventory

File	Purpose
<code>prompt.Rmd</code>	Slide generation prompt: instructs AI to extract all textbook examples from a given chapter and render them with dual mathematical/R solutions
<code>ex-review1.Rmd</code>	Automated grading prompt for Statistics (1) exercises
<code>ex-review2.Rmd</code>	Automated grading prompt for Statistics (2) exercises
<code>stat-system-guide.Rmd</code>	This technical document

8.1 Slide Generation Prompt (`prompt.Rmd`)

Key directives encoded in the slide generation prompt:

- **Role:** University-level statistics instructor.
- **Exhaustive extraction:** Every example in the target chapter must be incorporated; summarization or omission is not permitted.
- **Dual-solution requirement:** Each example must present (1) a full algebraic derivation in LaTeX and (2) executable R code with displayed output.
- **Paraphrasing mandate:** All prose must be rewritten in original language; verbatim textbook reproduction is prohibited.
- **Callout block usage:** Definitions, key formulae, and example prompts must be enclosed in appropriate callout blocks.
- **Directory replication:** Output must mirror the established file structure (e.g., `ch3/ch3.qmd`).

8.2 Grading Prompts (`ex-review1.Rmd`, `ex-review2.Rmd`)

Key components of the grading prompts:

- Definition of environment variables: `[no]`, `timezone (UTC+8)`, and root directory structure.
- Specification of all four grading stages with precise conditional logic.
- Output format requirements: scoreboard column names, Rmd content sections, and PDF compilation.

9 Operations Guide

9.1 New Semester Initialization

1. Export official class roster from the university administrative system.
Save to `stu1/` (Statistics 1) or `stu/` (Statistics 2).
2. Update `sylla/stat1.Rmd` and `sylla/stat2.Rmd` with the current semester's schedule and policies. Knit to PDF.
3. Verify `_quarto.yml` navigation links and semester metadata.
4. Rebuild the full website:
`quarto render`

9.2 Weekly Problem Set Publication

1. Create `stat1-ps/exN.qmd` (or `stat2-ps/exN.qmd`) in Reveal.js format.
Each slide corresponds to one question.
2. Render the problem set:
`quarto render stat1-ps/exN.qmd`
3. Update the problem set index in `prob.qmd`.

9.3 Weekly Submission Collection and Grading

1. Students submit to: `stat1-ex/exN/exN-[StudentID].pdf`
2. Execute the AI grading prompt (`prompt/ex-review1.Rmd`) with environment variable `[no]` set to the exercise number.
3. Verify generated scoreboard: `stu1/exN/ex-score.xlsx`
4. Verify analytics report: `stu1/exN/stat.pdf`

9.4 Generating a New Lecture Chapter

1. Duplicate the previous chapter directory:
`cp -r ch2/ ch3/`
2. Generate content using `prompt/prompt.Rmd` with an AI assistant, referencing the corresponding textbook chapter.
3. Review, edit, and finalize `ch3/ch3.qmd` in RStudio.
4. Render:
`quarto render ch3/ch3.qmd`
5. Add the chapter link to `slide.qmd`.

9.5 Full-Site Publication (GitHub Pages)

```
# From the stat/ root directory
quarto render

# Commit and push to trigger GitHub Pages deployment
git add .
```

```
git commit -m "Update: Week N - [brief description]"
git push origin main
```

9.6 Troubleshooting Reference

Table 10: Common Issues and Resolutions

Symptom	Probable Cause	Resolution
PDF output fails	TinyTeX not installed or incomplete	Run: <code>tinytex::install_tinytex()</code>
UUID package not found	uuid package not installed	Run: <code>install.packages('uuid')</code>
Quarto render error	Quarto version outdated	Update Quarto to latest stable release
Roster file read error	Legacy .xls format compatibility	Use readxl package or resave as .xlsx
Slide math not rendering	KaTeX configuration missing	Set <code>html-math-method: katex</code> in YAML

10 Course Chapter Reference

10.1 Statistics (1): Chapter Schedule

Table 11: Statistics (1) Weekly Schedule and Exercise Mapping

Week	Chapter	Topic	Exercise
2	Ch. 1	The Nature of Probability and Statistics	ex1
3	Ch. 2	Frequency Distributions and Graphs	ex2
4	Ch. 3	Data Description	ex3
5	Ch. 3	Data Description (continued)	ex4
6	Ch. 4	Probability and Counting Rules	ex5
7	Ch. 4	Probability and Counting Rules (continued)	ex6
8	—	Midterm Review	—
9	—	Midterm Examination	—
10	Ch. 5	Discrete Probability Distributions	ex7
11	Ch. 5	Discrete Probability Distributions (continued)	ex8
12	Ch. 6	The Normal Distribution	ex9
13	Ch. 6	The Normal Distribution (continued)	ex10
14	Ch. 7	Confidence Intervals and Sample Size	ex11
15	Ch. 7	Confidence Intervals and Sample Size (continued)	ex12
16	—	Final Examination	—

10.2 Statistics (2): Chapter Schedule

Table 12: Statistics (2) Weekly Schedule and Exercise Mapping

Week	Chapter	Topic	Exercises
2-4	Ch. 8	Hypothesis Testing	ex1-3
5-6	Ch. 9	Testing the Difference Between Two Means, Two Proportions, and Two Variances	ex4-5
7	Ch. 10	Correlation and Regression	ex6
8	—	Midterm Review	—
9	—	Midterm Examination	—
10-11	Ch. 10	Correlation and Regression (continued)	ex7-8
12-13	Ch. 11	Other Chi-Square Tests	ex9-10
14-15	Ch. 12	Analysis of Variance	ex11-12
16	—	Final Examination	—

11 Appendix

11.1 A. Official Submission Template

```

---
title: "Ex. 1"
author: "A113290001"
date: "2026-03-04"
output: pdf_document
---

# This chunk is mandatory in every submission.
print(uuid::UUIDgenerate())

# Exercise solutions follow below.

```

11.2 B. Directory Quick Reference

Table 13: Directory Purpose Quick Reference

Directory	Purpose
ch2/-ch14/	Reveal.js lecture slide sources, one subdirectory per chapter
stat1-ps/	Statistics (1) in-class exercise problem sets
stat2-ps/	Statistics (2) in-class exercise problem sets
stat1-ex/exN/	Statistics (1) student PDF submissions for exercise N
stat2-ex/exN/	Statistics (2) student PDF submissions for exercise N
stu1/exN/	Statistics (1) grading outputs (scoreboard, analytics report)
stu/exN/	Statistics (2) grading outputs (scoreboard, analytics report)
sylla/	Syllabus R Markdown source files
prompt/	AI-assisted prompt documents for slide generation and grading

11.3 C. References

- Bluman, A. G. (2012). *Elementary Statistics: A Step-by-Step Approach* (8th ed.). McGraw-Hill.
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- Course website: <https://yyliou.github.io/stat>